THOMPSON RIVERS UNIVERSITY

Department of Engineering

Faculty of Science

SENG 4660 Agile Game Development

Project Assignment: Design and Implementation of a Computer Game using Agile Process

Term: Winter 2026
Total Marks: 100

This is a group project with a maximum of two members.

Project Due: Last lab class (April 10th)

Student Learning Outcome:

LO-1 Understand the underlying concepts of agile process methodology in software engineering.

LO-2 Apply scrum process, identify sprints and generate events in game development projects.

LO-3 Analyze user stories, business rules, customer feedback, quality requirements.

LO-4 Develop agile plan and manage iterations in a multiple software releases.

LO-5 Write small scale computer game following agile approach.

Objective

Develop a small playable game using **any game engine** (Godot, Unity, Unreal, etc.) while **applying Agile principles**. Your goal is to demonstrate iterative development, teamwork, and effective use of Agile artifacts.

Minimum Game Requirements

Your game must include:

1. **Playable Character:** Player can control at least one character.
2. **Game Objective:** Clear goal or challenge (collect items, reach endpoint, defeat enemies).
3. **Interactions:** At least two interactive elements (collectibles, doors, enemies, power-ups).
4. **UI / HUD:** Display key info (**score**, health, timer, etc.).
5. **Feedback:** Visual, audio, or particle effects for actions/events **(Optional/Bonus)**.
6. **Agile Documentation:**
   - Product backlog
   - Sprint board (Kanban or Scrum board)
   - Retrospectives for each sprint

Agile Development Requirements

- Divide development into **4–5 sprints**.
- Each sprint must include:
  - **User stories:** Example format:

    "As a player, I want to collect gems so that my score increases."

  - **Sprint goal:** Example: "Implement player movement and basic controls."
  - **Task tracking:** Each task should be on a card in a Kanban board.
  - **Retrospective:** Complete a short reflection after each sprint.

Deliverables

1. **Playable Game** demonstrating required features.
2. **Git Repository / Version History** showing iterative development.
3. **Project Report** including:
   - Game concept & design
   - Screenshots or diagrams of backlog and sprint board
   - Retrospective notes
   - Sprint review summary
   - Daily scrum notes
   - Short explanation of tools/engine used
4. **Demo Presentation** (5–10 min) to the class.
5. **Video demo of your game (1-3 min maximum)**
6. **Final code as Appendix in the report**

Evaluation Criteria

Category	Weight	Details
Gameplay & Mechanics	35%	Game is functional, meets requirements, and is playable.
Agile Process & Documentation	30%	Clear backlog, sprint board, user stories, and retrospectives.
UI/HUD & Feedback	15%	HUD is clear; visual/audio feedback is meaningful.
Creativity & Originality	10%	Unique theme, mechanics, or design.
Presentation & Report	10%	Clear communication of process, decisions, and results.

Timeline & Milestones (Partial Submissions)

Milestone	Due
Project Proposal (Concept + Backlog)	Feb 20
Sprint 1 Demo & Retrospective	March 6
Sprint 2 Demo & Retrospective	March 20
Sprint 3 Demo & Retrospective	April 3rd
Playable game	April 3rd
Final Game & Report Submission	April 10
Final Presentation & Demo	April 10

Note: All partial submission should be done through GitHub

Sample Documents

Example User Stories

Feature	User Story
Player Movement	"As a player, I want to move my character so I can explore the map."
Collectibles	"As a player, I want to collect gems to increase my score."
Enemy AI	"As a player, I want to avoid enemies so that my health doesn't decrease."
HUD	"As a player, I want to see my health and score on screen."
Game Over	"As a player, I want the game to end when health reaches zero."

Sprint Board Example (Kanban)

To Do	In Progress	Done
Implement player movement	Add collision with walls	Player moves smoothly
Create collectibles	Script gem collection	Gems increase score
Add enemy AI	Script patrol behavior	Enemy moves and interacts

Update your sprint board after each lab or sprint. Include it in your project report.

Retrospective Template

Sprint #: _______
Date: _______
Team Members: _______

1. What went well ✅

- ____________
- ____________

2. What didn't go well ❌

- ____________
- ____________

3. What can be improved 💡

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4. Action items / Next Steps 📝

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Tips for Success

- Commit code frequently to Git to show incremental progress.
- Keep your user stories and tasks **small and achievable** per sprint.
- Playtest early and often. Feedback is part of Agile!
- Use visuals (diagrams, screenshots) to document progress.
- Be creative but ensure you meet **minimum requirements**.